TANKSHIFT – APPLICATION ANSWERS

Part 11 of the Master Pack – MBRDI × T-Hub Climate Tech Incubator: Student Launchpad Program, Season 2

Team: [Name 1], [Name 2], [Name 3] – REVA University

Date: 27 October 2026

11.1 What the Programme Page Says (checked 27 Oct 2026)

Item	Detail
Applications open / close	1 October 2026 / 15 November 2026
Cohort announced	21 December 2026
Launchpad runs	5 January – April 2027 (12 weeks)
Teams selected	18–20
Grant	Up to ₹50,000 per team, reimbursed against verified prototype milestones
Evaluation criteria	Problem–solution alignment; innovation potential; team strength; social impact potential; technical viability
Commitment required	Mentorship, workshops, maker-lab build days, weekly check-ins, Demo Day
Eligibility	Enrolled students, Karnataka (Bengaluru) and Maharashtra (Pune); cross-disciplinary teams encouraged
Contact	Czarina.arora@t-hub.co

Three things this changes in the pack:

- The 90-day pilot now runs inside the Launchpad, January–April 2027, not before. The bootcamp on 2 Nov is the pitch; the build comes after selection.
- The grant is reimbursed against milestones, not paid up front. The budget below is already split into milestones so the form answer matches how the money actually flows.
- The form link on the page is not live yet. Check the T-Hub page again on 1 November and submit well before 15 November.

11.2 Draft Answers (paste into the matching box; trim to any word limit)

Each answer is written to be read in under a minute by a reviewer scoring on the five criteria above. The word count is given so you can cut if the form has a limit.

Idea / startup name

TankShift

(1 words)

One-line description (tagline)

India’s water tanks are already a battery. We just switch it on.

(12 words)

Theme

Climate Intelligence & Energy Resilience – green energy and grid optimisation.

(11 words)

Describe the problem you are solving. Who faces it?

Every campus, hospital, tech park and factory pumps thousands of litres of water a day into overhead tanks and uses it hours later. The pumps run on float switches, blind to the clock. Karnataka’s compulsory Time-of-Day tariff for HT consumers above 500 kVA charges ₹1 extra per unit from 6–9 am and 6–10 pm and refunds ₹1 per unit from 10 pm–6 am (December–June). Demand charges of ₹350–380 per kVA penalise any pump that starts during the site’s peak. Facility teams have no affordable tool to act on either. The grid pays too: evening peaks are met by the dirtiest, costliest plants. Who feels it: universities, hospitals, IT parks, hotels, factories – and the DISCOM that buys the evening power.

(120 words)

Describe your solution.

TankShift is a low-cost retrofit controller that plugs into a site’s existing pump starter. It reads tank and sump levels, holds the local Time-of-Day tariff table and the site’s demand profile, and fills the tanks in the cheap night window while locking out the surcharge windows. A hard-coded reserve level guarantees water never runs out: if the tank drops below reserve it pumps at any hour. A 3-phase energy meter records every unit so the saving is proved on the actual electricity bill. No lithium, no civil work, nothing to replace: the tank is the battery, the pump is the charger, and we only change the timing.

(107 words)

What is innovative / unique about it? What exists today?

Pump timers exist but are blind to tariffs, levels and demand. Building-management systems (Schneider, Honeywell, Siemens) can do this but cost ₹10–50 lakh and are found only on sites that already have energy managers. Home batteries cost ₹1–5 lakh and degrade. No Indian product ties building pumping to Time-of-Day pricing, protects a water reserve, measures the saving on the bill, and can later aggregate many sites into a demand-response offer for the DISCOM. TankShift is the ₹10,000 version of a BMS feature for the 500 kVA–5 MVA sites nobody serves. The firmware is tariff-table driven, so the same box works in Maharashtra’s solar-hour rebate or any state’s future ToD order. We checked the idea against the KERC Electricity Tariff-2026 order line by line.

(123 words)

Technical feasibility – how will you build it? What is the prototype?

The prototype is a weatherproof box with an ESP32 microcontroller, a DS3231 real-time clock, a 3-phase Modbus energy meter with CTs, a relay that drives the existing starter’s Auto circuit, four float switches and an ultrasonic level sensor. Mains and logic sit in separate halves behind an insulating barrier; the electrician handles the panel side, we handle the low-voltage side. Firmware is a nine-state machine with a 10-second decision loop and eight hard-coded safety rules (reserve, dry-run, full-stop, sensor disagreement, boot-safe relay). Circuit, parts list, pin map, enclosure layout, firmware logic and a bench-test checklist are already written. A tabletop demo (fake clock, 12 V pump, phone page) exists to show the logic working. Everything uses parts available in SP Road, Bengaluru, for about ₹5,500 per HT-grade kit.

(128 words)

What is the impact? (environmental, social, economic)

Environmental: every unit moved off the 6–10 pm peak is a unit not generated by a peaking plant; it also helps the grid absorb more solar without new battery storage. One campus moves about 1,000–1,500 units a day. Economic: an estimated ₹2–5 lakh a year for a typical campus (base case about ₹3 lakh, from Time-of-Day energy and demand charges), from a ₹10,000 device – payback in weeks. The pilot replaces the estimate with the figure on REVA's own bill. The DISCOM gets flexible load without capital cost. Social: universities and hospitals first – institutions that pay the public's money for power. Residents and staff change nothing; water is always there. Circular: zero new storage material, zero end-of-life battery waste.

(120 words)

Market / who will pay?

Karnataka: every HT-2 consumer with contract demand ≥ 500 kVA is a compulsory ToD customer – universities, hospitals, IT parks, factories, hotels, malls. India: 20+ states have ToD for HT consumers; Maharashtra already pays solar-hour rebates. Water boards such as BWSSB (₹75 crore a month electricity bill) are a later enterprise segment. Residential becomes a market when domestic ToD arrives (central deadline April 2028). Who pays: the facility buys the hardware and an annual licence; later, the DISCOM pays a demand-response fee for aggregated load.

(85 words)

How will you use the ₹50,000 grant? (milestones)

Milestone 1 – Bench-proven kits (weeks 1–3, ₹14,000): 5 HT-grade controller kits built and tested on a lamp load at the REVA maker lab. Milestone 2 – Installed and logging (week 4, ₹12,000): 3 pump sets on REVA campus and 2 at a partner institution, installed with an electrician, logging to cloud. Milestone 3 – Measured savings (weeks 5–12, ₹10,000): 8 weeks of before/after data, SIM/cloud costs, site visits, bill collection. Step 2 (₹14,000, released only if Milestone 3 shows savings on the ToD bill): 2 further sites and a facility dashboard. Unused money returned. Output: a pilot report with measured savings from REVA's own ToD-slotted HT bill, ready to show BESCOM.

(112 words)

Why your team?

[Name 1] – hardware and electrical: builds the kits and works with the campus electrician. [Name 2] – firmware and app: ESP32, Modbus, the web dashboard. [Name 3] – sites, data and pitch: REVA estate office, partner institution, bill analysis, BESCOM. We are students on the very campus that is our first customer: a private university on HT-2(c) with compulsory ToD. We can reach the pump room, the estate office and the electricity bill in a day. [Add: relevant coursework, projects, hackathons, or faculty mentor.]

(85 words)

What have you done so far?

Scored 16 ideas against a simulated judge panel and killed the ones that already exist (second-life battery cooling, e-waste kiosks, cooling vests). Stress-tested TankShift against the KERC tariff order and corrected the target market from residential (no ToD in Karnataka) to HT consumers (compulsory ToD). Completed: 9-slide pitch, report, circuit design with parts list and pin map, 2D blueprint, 3D enclosure layout, firmware logic with state machine and code skeleton, tabletop demo design. Next: pull REVA's last six electricity bills to size the saving on real numbers.

(87 words)

What support do you need from the programme?

Maker-lab access for enclosure and panel work; a mentor with DISCOM or energy-audit experience; an introduction to the MBRDI Bengaluru campus facilities team as a second pilot site; guidance on BIS/CE-style compliance for a mains-connected device before any commercial sale.

(40 words)

How does this connect to mobility / MBRDI?

Stage 3 of the roadmap puts EV charger scheduling on the same controller: a campus or office fleet charges in the rebate window, the same way the tanks fill. Load shifting at the building edge is the cheapest form of grid integration for electric mobility.

(45 words)

Risks and how you handle them

Tank runs dry – hard-coded reserve, worst case it behaves like the float switch it replaced. Night pumping creates a new demand peak – staggered starts and a cap on simultaneous pumps. Monsoon months have no ToD adder – demand-charge logic and grid-friendly scheduling still apply. Site already has a BMS – not our customer. Tariff tables change – the table is a downloadable file, not firmware.

(67 words)

Commitment

All three members commit to the January–April 2027 Launchpad: workshops, maker-lab build days, weekly check-ins and Demo Day. The pilot site is our own campus, so build and measurement happen where we already are.

(34 words)

11.3 Before You Submit

- Fill in the three names, branches, years and phone numbers; add a faculty mentor if the form asks.
- Get one sentence of permission from the REVA estate office that the pump room can be used as a pilot site. A line in the form saying “pilot site confirmed” scores on technical viability.
- Attach the 9-slide deck as PDF if the form allows attachments. Attach the circuit and blueprint drawings.
- If there is a video field: 60 seconds of the tabletop demo (Part 10 script) on a phone is enough.
- Check the T-Hub page on 1 November for the live form link; submit by 10 November to leave margin before the 15 November close.
- Keep every answer under the limit shown; cut examples before cutting numbers.

Sources

T-Hub programme page: MBRDI Climate Tech Incubator – Student Launchpad Program Season 2 (t-hub.co), checked 27 Oct 2026. KERC Electricity Tariff-2026 (ToD tables, HT-2, demand charges). Earlier Master Pack sources for market and rejected ideas.